Prince Gupta

Senior Software Engineer | Backend Systems (Node.js, TypeScript, NestJS) | Gurgaon, India
+91-8059258639 | pg150704@gmail.com | LinkedIn | GitHub

Summary

Software Engineer with 3+ years of experience building production backend systems, APIs, integrations, and AI-enabled workflows using Node.js, TypeScript, and NestJS. Experienced in production debugging across Redis, background queues, OTP/SMS integrations, database performance, migrations, and Shopify synchronization. Strong in investigating failures end-to-end, learning unfamiliar systems quickly, and improving reliability through practical engineering.

Technical Skills

Backend: Node.js, NestJS, Express.js, REST APIs, GraphQL, microservices
Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis, indexing, query optimization
Messaging & Systems: BullMQ, asynchronous processing, event-driven workflows, retries, idempotency, failure recovery
Cloud & Delivery: AWS (EC2, RDS, S3, Lambda), Docker, Kubernetes, Jenkins, CI/CD
Languages: JavaScript, TypeScript, SQL, Python, Java, C++
AI / GenAI: LLM integration, AI-assisted workflows, content summarization, intelligent automation; exploring RAG and AI agents
Frontend: React.js, Next.js, HTML5, CSS3, API integration
Engineering: System design, distributed systems, concurrency, networking, observability, production debugging

Experience

Senior Software Engineer May 2023 – Present
Tech Almond Solutions Pvt. Ltd. (Almonds.ai) *Gurgaon, India*

- Build and maintain production services using Node.js, TypeScript, and NestJS for enterprise loyalty, engagement, and e-commerce platforms.
- Design REST and GraphQL APIs, reusable service modules, data models, and integrations for business-critical workflows.
- Build asynchronous workflows using queues, retries, and failure handling so unreliable third-party services do not unnecessarily block request processing.
- Improve database access and API responsiveness through SQL tuning, indexing, Redis caching, and efficient data-access patterns.
- Contribute to configurable, multi-tenant backend modules supporting different client workflows without duplicating core logic.
- Integrate Shopify APIs, payment gateways, OTP/SMS providers, IVR platforms, and other enterprise services.
- Collaborate on Docker, AWS, Jenkins, and Kubernetes workflows; investigate incidents across queues, Redis, databases, and integrations.
- Implement LLM/GenAI-enabled workflows for loyalty-platform automation, including AI-assisted processing and content summarization.
- Develop React-based product features and connect user-facing workflows with authenticated backend APIs.

Production Challenges

- **Background jobs not processing:** Investigated queue incidents across job state, worker execution, Redis connectivity, and retry behavior to distinguish enqueueing problems from worker-side failures.
- **Retry safety and duplicate events:** Considered duplicate delivery and repeated processing in queue and webhook workflows; use idempotency controls and state validation where implemented to prevent duplicate business actions.
- **Redis Cluster key-slot errors:** Investigated `CROSSSLOT` failures in multi-key operations by tracing key patterns and Cluster slot constraints; evaluated hash-tag-aware key design for operations that require related keys in the same slot.
- **Concurrent reward allocation:** Worked with Redis-backed reservation counters and reward limits, where concurrent requests require careful coordination to avoid exceeding campaign allocations.
- **Slow database queries:** Analysed query access patterns and indexes, then optimized query patterns and indexing to reduce execution time and unnecessary database work.
- **Incomplete database migration:** Compared source and destination schemas and table inventories after several tables were missed; validated migration completeness rather than relying only on the migration command status.
- **Third-party service failures:** Built or maintained asynchronous integration workflows with retries and failure handling so transient provider issues can be investigated and recovered without unnecessarily blocking user-facing requests.
- **Shopify webhook reliability:** Worked on failure logging and reprocessing workflows for synchronization events, accounting for retries and duplicate webhook delivery.
- **OTP and SMS inconsistencies:** Debugged flows across request validation, OTP expiry/state in Redis, queue processing, environment configuration, and provider responses to isolate application failures from delivery-provider issues.
- **LLM output reliability:** Integrated LLM-powered automation into loyalty workflows; validate and handle model responses at the application boundary before using them in downstream business processes.

Selected Projects

GPI – Loyalty & Productivity Platform | *Node.js, TypeScript, NestJS, Redis, AWS, Kubernetes*

- Built backend modules for loyalty and productivity workflows used by field users and retailers.
- Developed OTP- and IVR-based authentication flows integrated with SMS gateways and verification workflows.
- Created reusable modules and configurable workflows for multi-client deployments; contributed to Redis-backed reward-processing logic.

Fills – Shopify-Integrated Marketplace | *Node.js, TypeScript, Shopify APIs, Redis, AWS*

- Built backend integrations for product, variant, inventory, and order synchronization between the marketplace and Shopify.
- Developed webhook-processing workflows with failure logging and recovery/reprocessing paths for unsuccessful events.

HCCB DMC – Driver Management System | *Node.js, MySQL, Redis, Docker, Firebase*

- Built backend functionality for Coca-Cola driver-management workflows and operational processes.
- Optimized SQL queries and indexing strategies to improve database execution performance.
- Implemented secure data exchange using asymmetric cryptography mechanisms.

Loyalty Platform – LLM / GenAI Automation | *Node.js, TypeScript, NestJS, LLM integration*

- Implemented LLM-enabled workflows in a loyalty platform to automate selected content-processing and summarization tasks.
- Integrated AI processing into backend workflows while keeping application services responsible for orchestration and business logic.
- Considered failure handling and operational fit when incorporating AI-assisted processing into existing workflows.

Current Learning

Deepening practical knowledge of React and full-stack architecture; RAG, embeddings, vector search, and AI-agent patterns; database internals and query execution; distributed systems, concurrency, networking, and production observability. Strengthening system design through the study of failure modes, scalability, reliability, and architectural trade-offs.

Education

Indira Gandhi University (IGU) — Haryana, India
Bachelor of Technology in Computer Science Engineering — *2019 – 2023*